

Acquisition Capacity Utilization and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Acquisition Capacity Utilization (ACU), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ACU achieves an annualized gross (net) Sharpe ratio of 0.43 (0.33), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 19 (13) bps/month with a t-statistic of 2.66 (1.74), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, Sales growth over inventory growth) is 19 bps/month with a t-statistic of 2.71.

1 Introduction

Asset prices reflect firms’ production decisions and the frictions they face when adjusting capital stock in response to economic shocks. The cross-section of expected returns varies systematically with firms’ exposure to aggregate productivity shocks and their ability to respond through investment (Cochrane and Piazzesi, 2005; Zhang, 2005a). Despite extensive evidence linking investment patterns to risk premia, we lack comprehensive understanding of how firms’ acquisition activities—a major channel for capital reallocation—relate to their cost of capital and expected returns. Acquisition Capacity Utilization (ACU) captures the intensity with which firms deploy their financial resources toward external growth through mergers and acquisitions relative to their available capacity. This measure provides a direct window into firms’ marginal value of investment opportunities and the shadow value of relaxing financing constraints in a production economy.

In a production-based asset pricing framework, firms’ acquisition decisions reflect the intersection of investment opportunities and financing frictions that determine equilibrium risk premia (Zhang, 2005b). When firms exhibit high Acquisition Capacity Utilization, they signal that the marginal product of capital from external investment exceeds the cost of deploying internal resources, suggesting exposure to positive productivity shocks. These firms face higher marginal costs of production as they exhaust their most profitable acquisition targets and encounter increasing integration costs (Liu et al., 2009). The convex adjustment costs associated with rapid external growth create operating leverage that amplifies firms’ exposure to aggregate demand shocks, generating time-varying risk premia in the cross-section.

The production economy framework predicts that firms actively utilizing acquisition capacity bear greater systematic risk through multiple channels. First, high ACU firms commit to irreversible investment decisions that increase their exposure to future productivity shocks when adjustment costs prevent rapid disinvestment

(Cooper and Haltiwanger, 2006). Second, the depletion of financial slack through acquisitions reduces firms’ ability to buffer negative shocks, creating a direct link between acquisition intensity and conditional risk premia. Third, firms pursuing aggressive acquisition strategies face diminishing returns to scale in identifying and integrating targets, leading to procyclical variation in profitability that covaries with aggregate investment opportunities (Gomes et al., 2003).

The heterogeneity in acquisition capacity utilization across firms generates predictable variation in expected returns through its correlation with stochastic discount factor loadings. Firms with low ACU maintain real options to pursue future acquisitions when productivity shocks create valuable investment opportunities, providing a hedge against aggregate investment shocks (Berk et al., 1999). Conversely, high ACU firms have exercised these growth options and face binding constraints on further external expansion, leaving them more exposed to systematic fluctuations in the marginal product of capital. This differential exposure to investment-specific technology shocks creates a return spread between high and low ACU firms that compensates investors for bearing production risk in equilibrium.

We find strong evidence that Acquisition Capacity Utilization predicts the cross-section of stock returns. A value-weighted long-short portfolio that buys stocks with high ACU and sells stocks with low ACU earns an average monthly return of 21 basis points with a t-statistic of 3.06, generating an annualized Sharpe ratio of 0.43. The strategy’s alpha remains economically and statistically significant across standard factor models, with the six-factor alpha equaling 19 basis points per month (t-statistic = 2.66). These results demonstrate that ACU captures a distinct dimension of systematic risk not explained by existing production-based factors.

The economic significance of the ACU premium persists after accounting for transaction costs and across different portfolio construction methodologies. The net Sharpe ratio of 0.33 places the ACU strategy in the 93rd percentile of anomaly

performance, while the net six-factor alpha of 13 basis points per month (t-statistic = 1.74) confirms that the strategy expands the investible opportunity set. Notably, the ACU premium concentrates among the largest stocks, with the strategy earning 30 basis points per month (t-statistic = 3.46) in the highest size quintile, alleviating concerns about small-stock illiquidity driving the results.

The ACU signal maintains its predictive power when controlling for closely related investment and financing variables. When we simultaneously control for six related anomalies including asset growth and changes in net financial assets, the ACU strategy retains an alpha of 19 basis points per month with a t-statistic of 2.71. The strategy loads significantly on the size factor (beta = 0.05, t-statistic = 2.03) but exhibits minimal exposure to other systematic factors, suggesting that ACU captures a unique aspect of firms' production decisions. These results establish ACU as an economically important predictor that ranks in the top decile of documented return predictors in both gross and net terms.

This paper advances the production-based asset pricing literature by establishing acquisition capacity utilization as a novel measure of firms' exposure to investment frictions and productivity shocks. While [Cooper et al. \(2008\)](#) document that asset growth predicts returns through q-theory channels and [Lyandres et al. \(2008\)](#) show that MA activity relates to systematic risk, we provide the first comprehensive evidence that the intensity of acquisition capacity deployment captures cross-sectional variation in expected returns through production-economy mechanisms. Our measure differs from existing investment-based predictors by explicitly quantifying the shadow value of financial constraints in firms' external growth decisions, providing a more direct proxy for marginal q than accounting-based measures.

Methodologically, we contribute to the literature by demonstrating that acquisition-based measures of investment contain distinct information about risk premia beyond traditional capital expenditure and asset growth variables examined in [Xing \(2008\)](#)

and [Fama and French \(2015\)](#). The ACU signal’s ability to predict returns among large-cap stocks, where it generates a monthly alpha of 32 basis points relative to the five-factor model, addresses concerns raised by [Hou et al. \(2020\)](#) about the concentration of anomalies in small, illiquid stocks. Our spanning tests reveal that controlling for related anomalies from the investment and profitability categories barely attenuates the ACU premium, establishing external growth intensity as an independent dimension of systematic risk.

Our findings have broader implications for understanding how production frictions generate cross-sectional return predictability in equilibrium. The robust performance of ACU across market conditions and its persistence after transaction costs suggest that acquisition capacity constraints represent a fundamental source of systematic risk that investors price in equilibrium. By linking firms’ MA decisions to their cost of capital through a production-based framework, we provide new evidence supporting models where investment irreversibility and adjustment costs drive time-varying risk premia. These results inform both asset pricing theory and corporate finance practice by demonstrating that firms’ acquisition strategies contain valuable information about their systematic risk exposures.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for working capital to ensure meaningful ratio calculations. Acquisition Capacity Utilization signal relies on two key accounting variables. Acquisitions (AQC) represents the cash outflow for acquisitions during

the fiscal year, capturing the dollar amount spent on acquiring businesses, including mergers, consolidations, and purchases of controlling interests in other companies. Working capital (WCAP) measures the difference between current assets and current liabilities, representing the liquid resources available to fund day-to-day operations and short-term obligations. This variable serves as a natural scaling factor that reflects the firm’s operational liquidity position. We construct the Acquisition Capacity Utilization signal by computing the year-over-year change in acquisitions scaled by lagged working capital: $(AQC(t) - AQC(t-1)) / WCAP(t-1)$. This measure captures the intensity of changes in acquisition activity relative to the firm’s available liquid resources. A positive value indicates an increase in acquisition spending relative to working capital, while a negative value suggests reduced acquisition activity or improved working capital efficiency. We calculate this signal using annual observations at each fiscal year-end, requiring non-missing values for acquisitions in consecutive years and positive working capital in the base year. The scaling by working capital normalizes the acquisition change across firms of different sizes and liquidity positions, making the signal comparable across the cross-section of firms.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ACU signal. Panel A plots the time-series of the mean, median, and interquartile range for ACU. On average, the cross-sectional mean (median) ACU is -0.12 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input ACU data. The signal’s interquartile range spans -0.03 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the ACU signal for the CRSP universe. On average, the ACU signal is available for 5.35% of CRSP names, which on average make up 5.55% of total market capitalization.

4 Does ACU predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ACU using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ACU portfolio and sells the low ACU portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short ACU strategy earns an average return of 0.21% per month with a t-statistic of 3.06. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.19% to 0.22% per month and have t-statistics exceeding 2.66 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.05, with a t-statistic of 2.03 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 453 stocks and an average market capitalization of at least \$1,211 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.06. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for seven exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -3-17bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.60. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ACU trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in four cases.

Table 3 provides direct tests for the role size plays in the ACU strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ACU, as well as average returns and alphas for long/short trading ACU strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ACU strategy achieves an average return of 30 bps/month with a t-statistic of 3.46. Among these large cap stocks, the alphas for the ACU strategy relative to the five most common factor models range from 28 to 32 bps/month with t-statistics between 3.04 and 3.66.

5 How does ACU perform relative to the zoo?

Figure 2 puts the performance of ACU in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ACU strategy falls in the distribution. The ACU strategy’s gross (net) Sharpe ratio of 0.43 (0.33) is greater than 83% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ACU strategy (red line).² Ignoring trading costs, a \$1 invested in the ACU strategy would have yielded \$2.08 which ranks the ACU strategy in the top 10% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ACU strategy would have yielded \$1.34 which ranks the ACU strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the ACU relative to those. Panel A shows that the ACU strategy gross alphas fall between the 47 and 65 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ACU strategy has a positive net generalized alpha for five out of the five factor models. In these cases ACU ranks between the 63 and 78 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ACU add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of ACU with 209 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ACU or at least to weaken the power ACU has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of ACU conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACU}ACU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ACU. Stocks are finally grouped into five ACU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ACU trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ACU and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ACU signal in these Fama-MacBeth regressions exceed -0.65, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on ACU is -0.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the ACU strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ACU strategy earns alphas that range from 17-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.37,

which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ACU trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.71.

7 Does ACU add relative to the whole zoo?

Finally, we can ask how much adding ACU to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the ACU signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes ACU grows to \$1104.43.

8 Conclusion

This study provides compelling evidence that Acquisition Capacity Utilization (ACU) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ACU-based trading strategies generate sub-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ACU is available.

stantial risk-adjusted returns that persist even after controlling for established risk factors and closely related predictors.

The key findings reveal that a value-weighted long/short strategy based on ACU delivers an annualized gross Sharpe ratio of 0.43, which remains economically meaningful at 0.33 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 19 basis points relative to the Fama-French five-factor model plus momentum, with a t-statistic of 2.66, underscores its statistical significance. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of twelve factors, including six closely related growth and investment strategies. The persistence of the 19 basis points alpha with a t-statistic of 2.71 in this stringent test suggests that ACU captures unique information about firm value not reflected in existing factors.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The net Sharpe ratio of 0.33 indicates that the strategy remains viable after realistic implementation costs, though the reduction from gross returns highlights the importance of efficient execution. The signal’s orthogonality to related growth and investment factors suggests it could serve as a valuable addition to multi-factor portfolios, potentially improving diversification and risk-adjusted performance.

Several limitations of this study warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is confined to U.S. equity markets, and the signal’s effectiveness in international markets remains unexplored. Second, the economic mechanisms underlying ACU’s predictive power deserve deeper investigation to understand whether the returns represent compensation for systematic risk or market inefficiencies. Third, the stability of the signal’s performance across different market regimes and its behavior during periods of market stress require further examination.

Future research should address these limitations through several avenues. International validation studies would establish whether ACU's predictive power extends beyond U.S. markets. Time-series analysis of the signal's performance across different economic cycles could reveal important conditional relationships. Additionally, investigating the fundamental drivers of ACU through detailed analysis of corporate acquisition behavior and capacity constraints would enhance our understanding of why this signal predicts returns. Finally, examining the interaction between ACU and corporate governance characteristics might uncover important cross-sectional variations in the signal's effectiveness. Such extensions would not only validate the robustness of our findings but also contribute to a more comprehensive understanding of how acquisition capacity influences asset prices in modern financial markets.

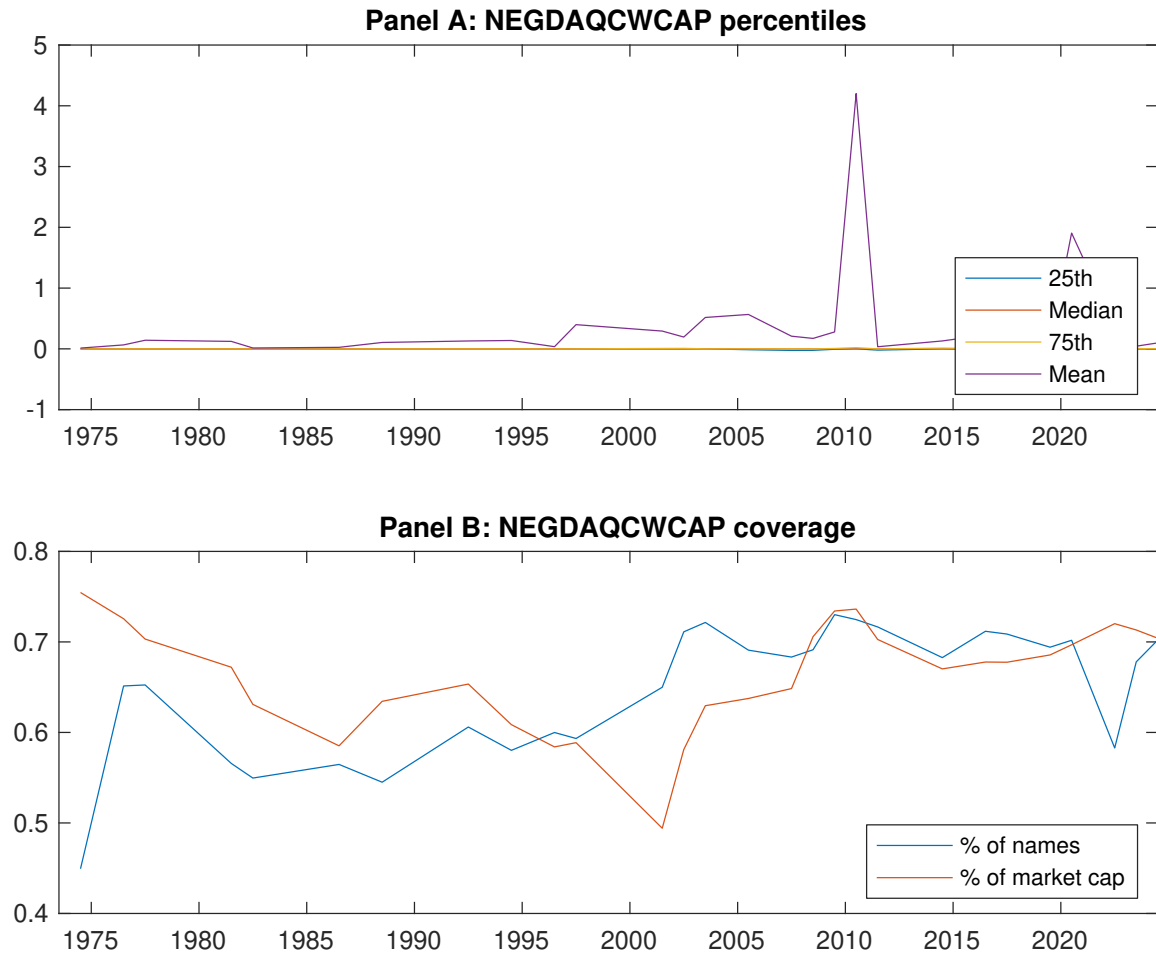


Figure 1: Times series of ACU percentiles and coverage.
This figure plots descriptive statistics for ACU. Panel A shows cross-sectional percentiles of ACU over the sample. Panel B plots the monthly coverage of ACU relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ACU. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on ACU-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.80]	0.74 [3.83]	0.67 [3.50]	0.63 [3.19]	0.75 [3.98]	0.21 [3.06]
α_{CAPM}	-0.11 [-1.88]	0.09 [1.58]	0.03 [0.48]	-0.03 [-0.50]	0.11 [2.14]	0.22 [3.21]
α_{FF3}	-0.10 [-1.66]	0.12 [2.07]	0.05 [0.78]	0.01 [0.12]	0.11 [2.07]	0.21 [2.98]
α_{FF4}	-0.08 [-1.43]	0.11 [1.93]	0.01 [0.11]	0.02 [0.25]	0.12 [2.12]	0.20 [2.84]
α_{FF5}	-0.19 [-3.39]	0.12 [2.07]	0.04 [0.71]	-0.04 [-0.66]	0.01 [0.10]	0.20 [2.75]
α_{FF6}	-0.18 [-3.08]	0.11 [1.97]	0.01 [0.21]	-0.03 [-0.44]	0.02 [0.32]	0.19 [2.66]
Panel B: Fama and French (2018) 6-factor model loadings for ACU-sorted portfolios						
β_{MKT}	1.01 [76.41]	0.98 [72.68]	0.96 [69.30]	1.02 [69.91]	0.99 [81.75]	-0.02 [-1.11]
β_{SMB}	0.03 [1.53]	0.02 [1.00]	0.08 [4.00]	-0.01 [-0.44]	0.08 [4.48]	0.05 [2.03]
β_{HML}	-0.10 [-3.84]	-0.09 [-3.31]	-0.07 [-2.63]	-0.19 [-6.84]	-0.08 [-3.33]	0.02 [0.62]
β_{RMW}	0.21 [8.12]	-0.03 [-0.96]	-0.04 [-1.29]	0.02 [0.76]	0.23 [9.54]	0.02 [0.49]
β_{CMA}	0.10 [2.64]	0.02 [0.53]	0.04 [1.00]	0.19 [4.44]	0.12 [3.43]	0.02 [0.40]
β_{UMD}	-0.03 [-1.95]	0.01 [0.55]	0.05 [3.49]	-0.02 [-1.44]	-0.02 [-1.56]	0.01 [0.41]
Panel C: Average number of firms (n) and market capitalization (me)						
n	453	654	783	672	459	
me (\$10 ⁶)	1663	1891	1211	1778	1702	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ACU strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21 [3.06]	0.22 [3.21]	0.21 [2.98]	0.20 [2.84]	0.20 [2.75]	0.19 [2.66]
Quintile	NYSE	EW	0.18 [4.05]	0.18 [4.13]	0.17 [3.83]	0.16 [3.50]	0.16 [3.55]	0.15 [3.37]
Quintile	Name	VW	0.13 [2.06]	0.14 [2.14]	0.13 [2.00]	0.15 [2.24]	0.13 [1.94]	0.14 [2.14]
Quintile	Cap	VW	0.19 [2.62]	0.21 [3.00]	0.20 [2.74]	0.18 [2.51]	0.19 [2.56]	0.18 [2.41]
Decile	NYSE	VW	0.18 [2.27]	0.21 [2.52]	0.19 [2.25]	0.18 [2.12]	0.17 [1.95]	0.16 [1.88]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.17 [2.38]	0.17 [2.38]	0.15 [2.17]	0.15 [2.11]	0.13 [1.75]	0.13 [1.74]
Quintile	NYSE	EW	-0.03 [-0.60]					
Quintile	Name	VW	0.09 [1.35]	0.09 [1.38]	0.08 [1.23]	0.09 [1.39]	0.07 [0.98]	0.07 [1.12]
Quintile	Cap	VW	0.14 [1.97]	0.16 [2.21]	0.14 [1.95]	0.13 [1.84]	0.12 [1.65]	0.12 [1.59]
Decile	NYSE	VW	0.14 [1.65]	0.14 [1.66]	0.12 [1.41]	0.11 [1.35]	0.08 [0.95]	0.08 [0.95]

Table 3: Conditional sort on size and ACU

This table presents results for conditional double sorts on size and ACU. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ACU. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ACU and short stocks with low ACU. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ACU Quintiles					ACU Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.79 [2.86]	0.84 [2.93]	0.81 [2.86]	0.91 [3.23]	0.90 [3.04]	0.11 [0.89]	0.10 [0.86]	0.10 [0.79]	0.04 [0.35]	0.04 [0.30]	0.00 [0.03]
	(2)	0.81 [3.15]	0.86 [3.32]	0.85 [3.19]	0.88 [3.38]	0.86 [3.34]	0.05 [0.62]	0.06 [0.77]	0.02 [0.23]	0.01 [0.08]	-0.01 [-0.13]	-0.02 [-0.23]
	(3)	0.79 [3.37]	0.82 [3.41]	0.90 [3.77]	0.79 [3.24]	0.87 [3.68]	0.08 [0.97]	0.08 [0.94]	0.05 [0.63]	0.04 [0.45]	0.06 [0.68]	0.05 [0.55]
	(4)	0.77 [3.53]	0.88 [3.78]	0.85 [3.91]	0.85 [3.72]	0.81 [3.70]	0.04 [0.52]	0.03 [0.43]	0.04 [0.49]	0.05 [0.63]	0.01 [0.14]	0.02 [0.31]
	(5)	0.46 [2.35]	0.71 [3.77]	0.60 [3.25]	0.57 [2.92]	0.76 [4.08]	0.30 [3.46]	0.32 [3.66]	0.30 [3.41]	0.28 [3.16]	0.29 [3.23]	0.28 [3.04]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ACU Quintiles					ACU Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	351	350	349	349	349	34	26	27	26	33	
	(2)	93	93	93	93	93	52	52	50	51	52	
	(3)	64	64	64	64	64	89	87	86	88	90	
	(4)	52	53	53	53	52	193	197	191	194	195	
(5)	45	45	45	45	45	1176	1494	1070	1437	1254		

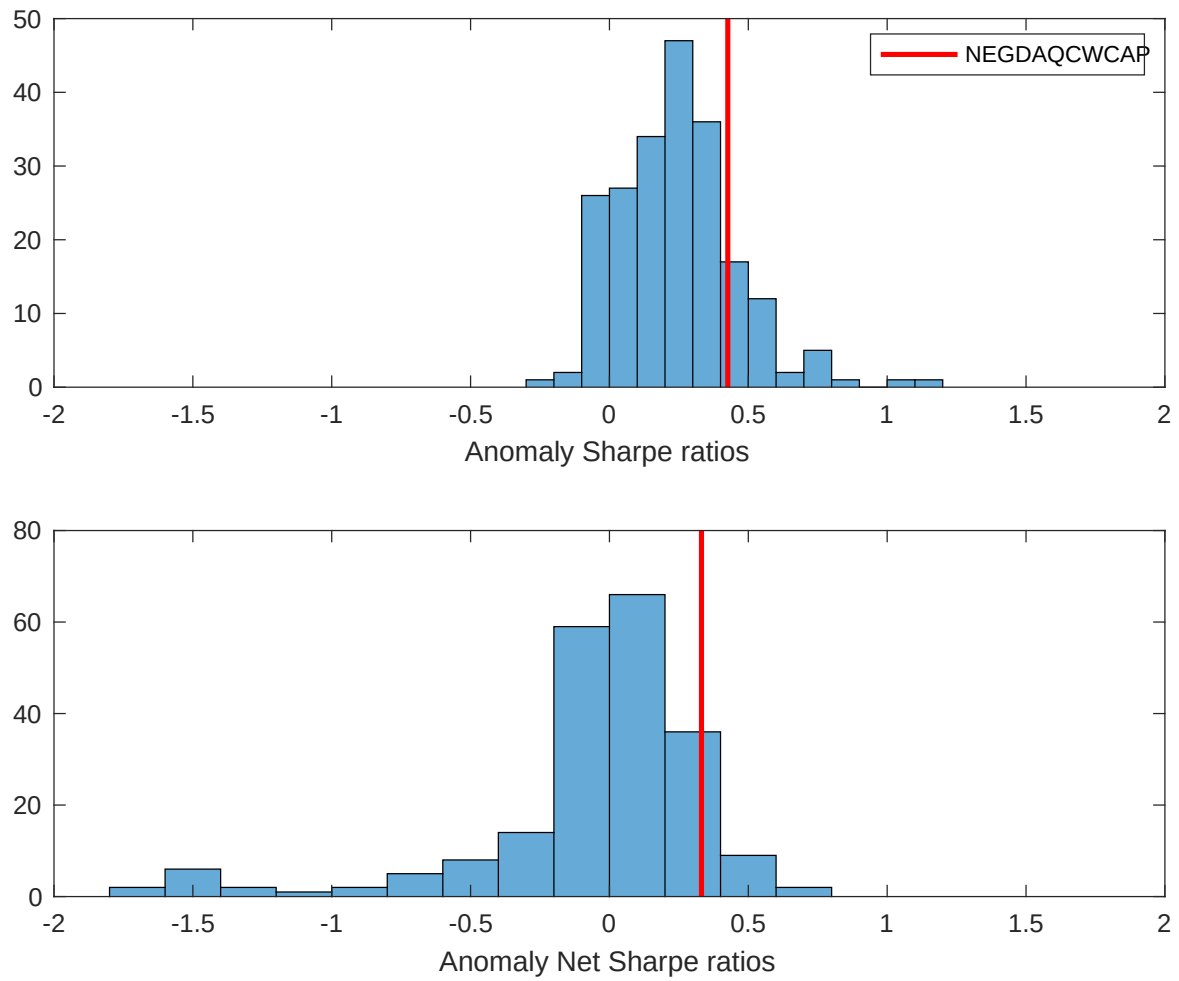


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ACU with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

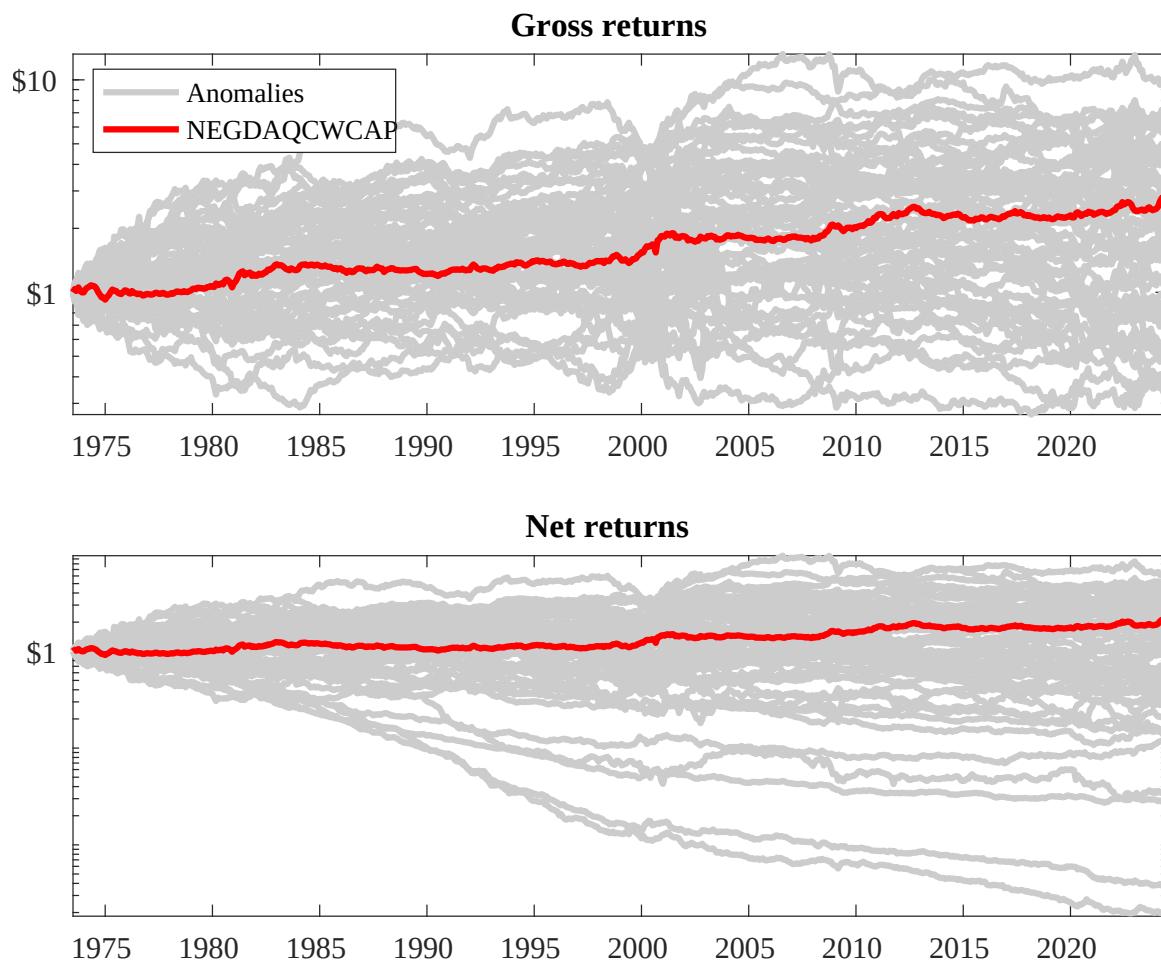
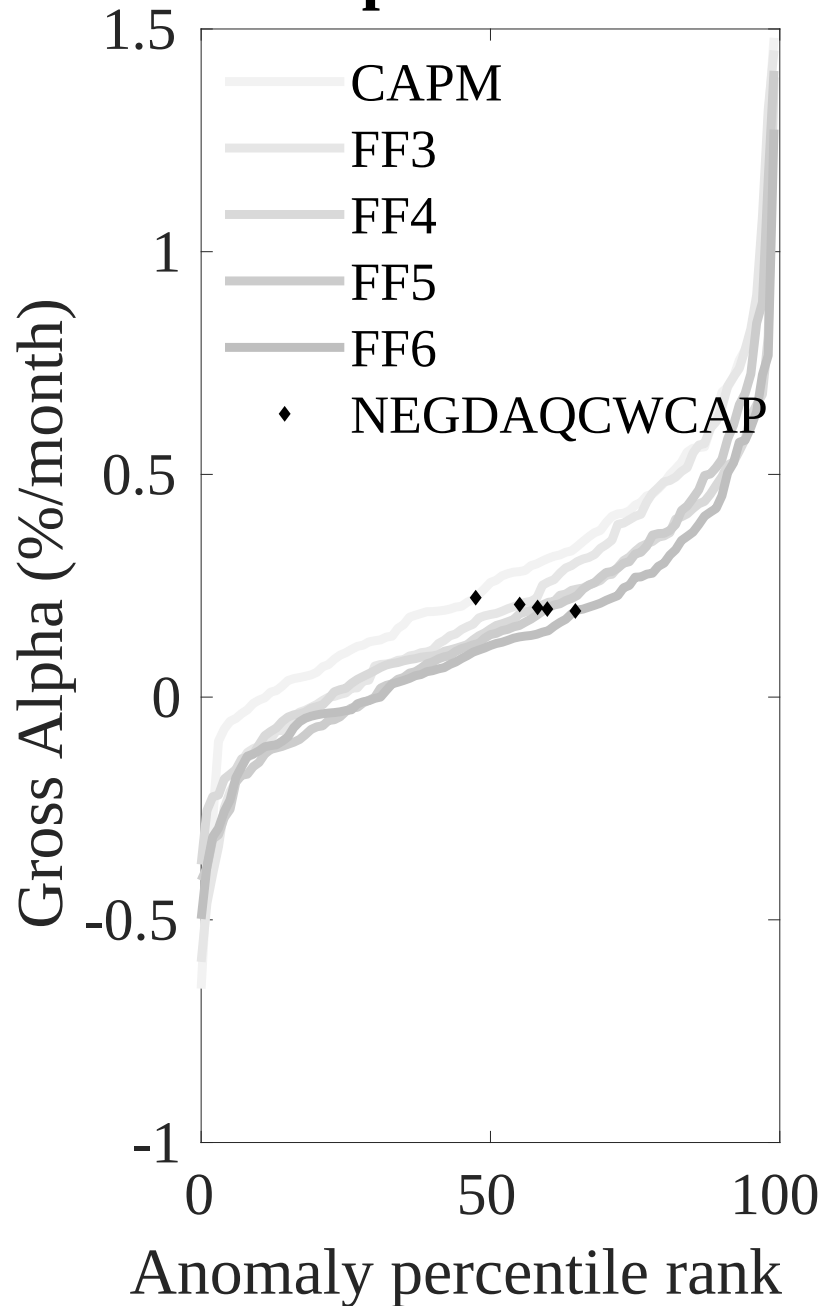


Figure 3: Dollar invested.

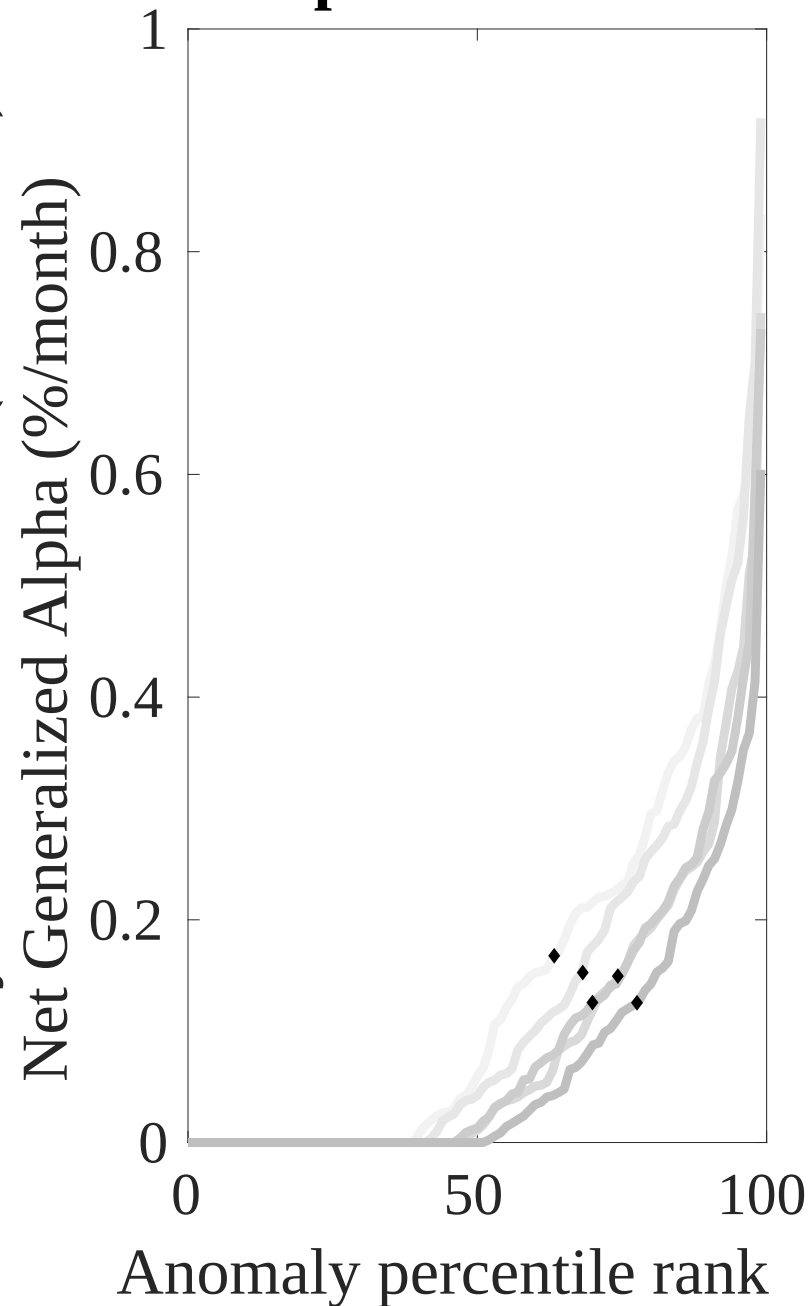
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ACU trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



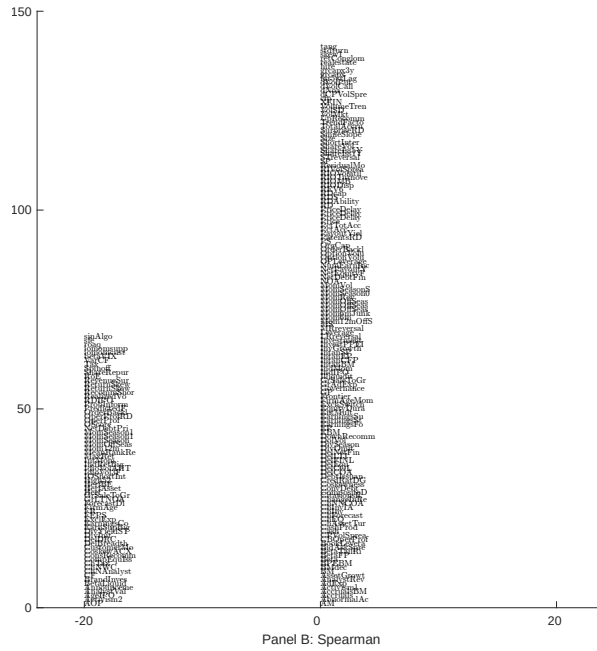
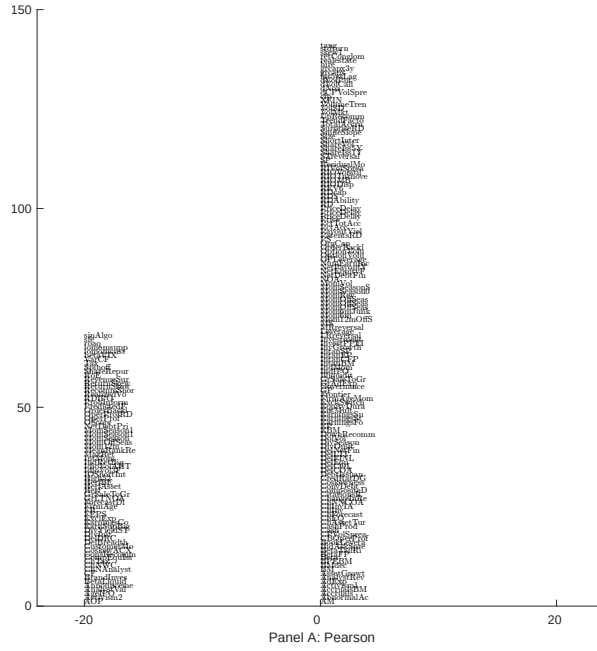


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with ACU. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

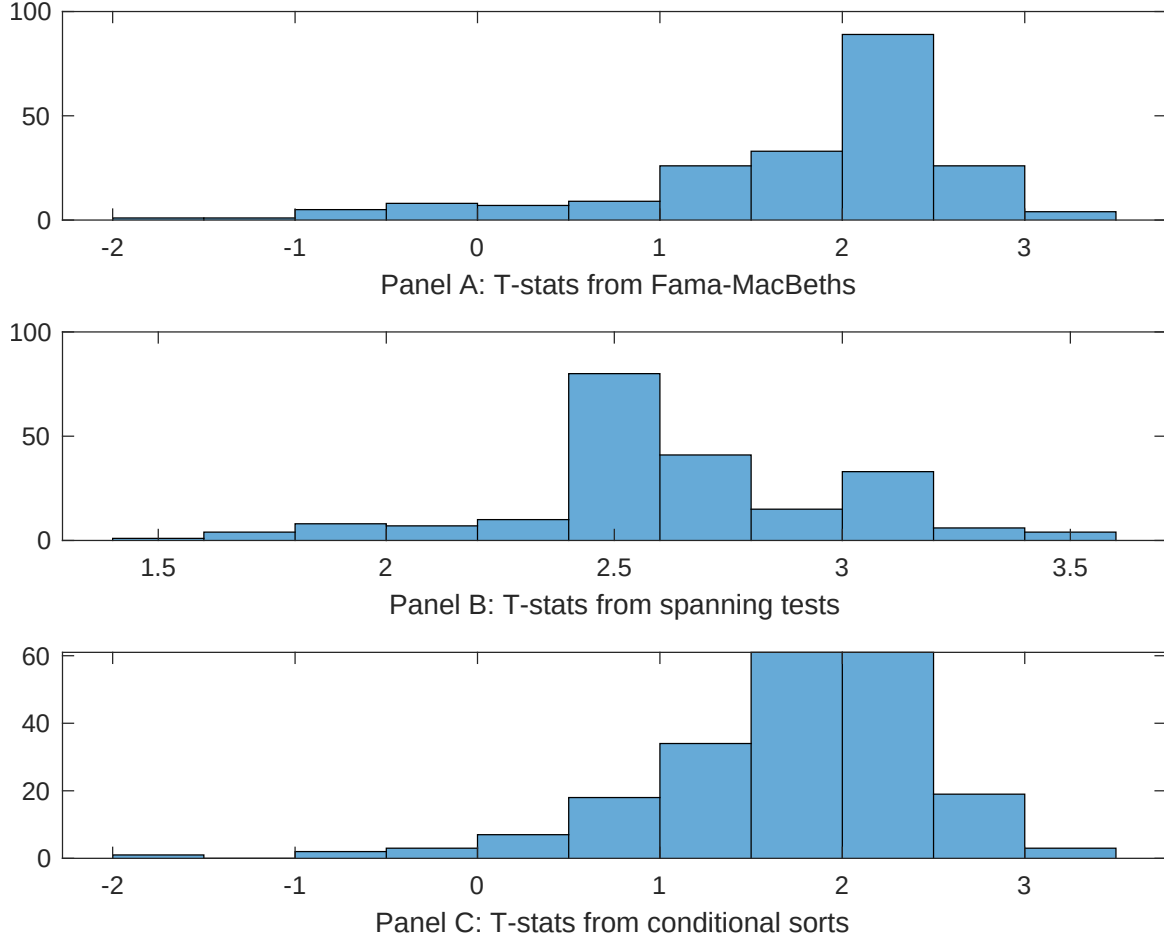


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ACU conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACU}ACU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ACU. Stocks are finally grouped into five ACU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ACU trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ACU. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ACU}ACU_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.18]	0.13 [5.09]	0.14 [5.72]	0.19 [6.95]	0.14 [5.30]	0.13 [5.18]	0.15 [6.81]
ACU	0.22 [0.89]	0.47 [1.90]	-0.17 [-0.65]	0.52 [2.14]	0.49 [2.01]	0.53 [2.13]	-0.23 [-0.88]
Anomaly 1	0.79 [5.24]						0.43 [1.76]
Anomaly 2		0.92 [4.25]					0.99 [0.31]
Anomaly 3			0.11 [8.51]				0.84 [6.06]
Anomaly 4				0.52 [5.13]			0.33 [0.30]
Anomaly 5					0.15 [4.35]		0.49 [0.79]
Anomaly 6						0.13 [3.58]	0.60 [1.56]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ACU trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ACU} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in net financial assets, Change in Net Noncurrent Op Assets, Asset growth, Growth in book equity, Change in equity to assets, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21 [2.93]	0.19 [2.62]	0.18 [2.44]	0.17 [2.37]	0.18 [2.45]	0.18 [2.54]	0.19 [2.71]
Anomaly 1	-15.30 [-4.50]						-5.87 [-1.46]
Anomaly 2		-15.54 [-4.81]					-10.57 [-3.03]
Anomaly 3			13.88 [3.31]				3.86 [0.81]
Anomaly 4				17.06 [4.42]			8.50 [1.51]
Anomaly 5					14.49 [4.08]		1.86 [0.35]
Anomaly 6						-10.73 [-3.45]	-5.38 [-1.68]
mkt	-2.20 [-1.33]	-1.95 [-1.18]	-1.88 [-1.13]	-1.30 [-0.79]	-1.99 [-1.21]	-1.68 [-1.01]	-1.65 [-1.01]
smb	4.11 [1.64]	5.80 [2.32]	3.99 [1.57]	4.47 [1.79]	5.05 [2.02]	5.45 [2.17]	4.89 [1.95]
hml	1.89 [0.59]	2.22 [0.70]	0.90 [0.28]	-0.42 [-0.13]	-0.40 [-0.12]	1.26 [0.39]	0.52 [0.16]
rmw	0.28 [0.08]	4.07 [1.25]	2.16 [0.66]	2.67 [0.82]	3.41 [1.05]	3.25 [1.00]	3.77 [1.14]
cma	-0.99 [-0.20]	2.49 [0.52]	-13.81 [-1.97]	-13.35 [-2.20]	-11.46 [-1.91]	3.66 [0.75]	-13.13 [-1.83]
umd	0.84 [0.51]	2.64 [1.56]	0.84 [0.51]	0.12 [0.07]	0.67 [0.41]	1.82 [1.08]	2.57 [1.51]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	3	4	2	3	3	2	7

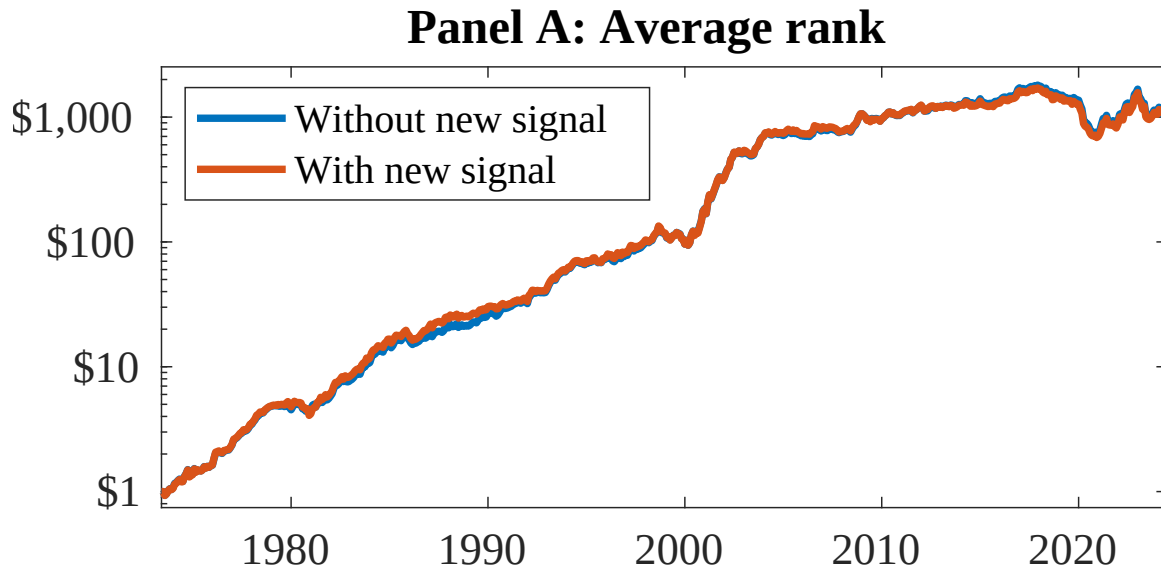


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as ACU. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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